PROJECT PROPOSAL · STRUCTURE & PLAN

Sri Lanka Share Market News Analysis Agent

A daily news-analysis tool for the Colombo Stock Exchange

PROJECT TYPE	AI / Machine Learning group project
MARKET	Colombo Stock Exchange (CSE), Sri Lanka
PHASE 1 GOAL	Daily sector-level news report
DURATION	Approximately 10 weeks
SUBMITTED TO	Project Supervisor

> **In one sentence.** This project reads Sri Lankan financial news every day, works out which market sector each story is about, and produces a short daily report showing which sectors received good or bad news, and why. It does *not* predict share prices.

1 Project Summary

1.1 The problem

A person following the Colombo Stock Exchange has to read many news websites every day. The same story appears on several sites. Some stories matter a lot and some do not matter at all. It is slow work, and it is easy to miss something important.

1.2 What we will build

A program that does this reading automatically. Every morning it produces one short report: the most important news of the last 24 hours, grouped by market sector, with a simple score showing whether each sector received good news or bad news.

1.3 What we will not build

We will not build a system that predicts share prices. This is a deliberate decision, and we explain the reason below because it shapes the whole design.

> **WHY WE DO NOT PREDICT PRICES**
> The Sri Lankan market is small. For any one type of news event there may be only 5 to 15 past examples. That is far too few to train a model that predicts price movements reliably.
>
> There is also a practical reason. A tool that says *“this share will rise 4%”* will be wrong often, and people will stop trusting it within a month. A tool that says *“this sector received 3 positive news items today, here they are”* is simply reporting facts, so it stays useful.
>
> **Our system presents evidence. The human makes the decision.**

1.4 Why this is still a useful project

- It saves time — one report instead of reading six websites.
- It never misses an announcement, which a tired human easily does.
- It removes repeated stories, so the same news is not counted five times.
- It builds a valuable dataset (see Section 6) that makes Phase 2 possible.
- It is honest about uncertainty, which most similar tools are not.

2 Scope

We limit the scope carefully so the project can be finished properly rather than half-finished.

IN SCOPE (PHASE 1)	OUT OF SCOPE (LATER OR NEVER)
Reading news from CSE and financial news websites	Predicting exact share prices or percentages
Deciding which **sector** each story belongs to (banks, construction, hotels, etc.)	Deciding which **exact company** each story is about
Deciding if news is good or bad, big or small	Buying or selling shares automatically
Removing duplicate stories	Real-time / minute-by-minute updates
A daily report with scores and reasons	Reading quarterly financial report PDFs (Phase 3)
Saving daily closing prices for future use	Giving personal investment advice

AN IMPORTANT SCOPE DECISION: SECTOR LEVEL, NOT COMPANY LEVEL

Working out which *exact company* a news article is about is the hardest part of this kind of system. The same company is written many ways — “Sampath”, “Sampath Bank PLC”, or the Sinhala name. Handling this properly needs a large hand-made list and weeks of work.

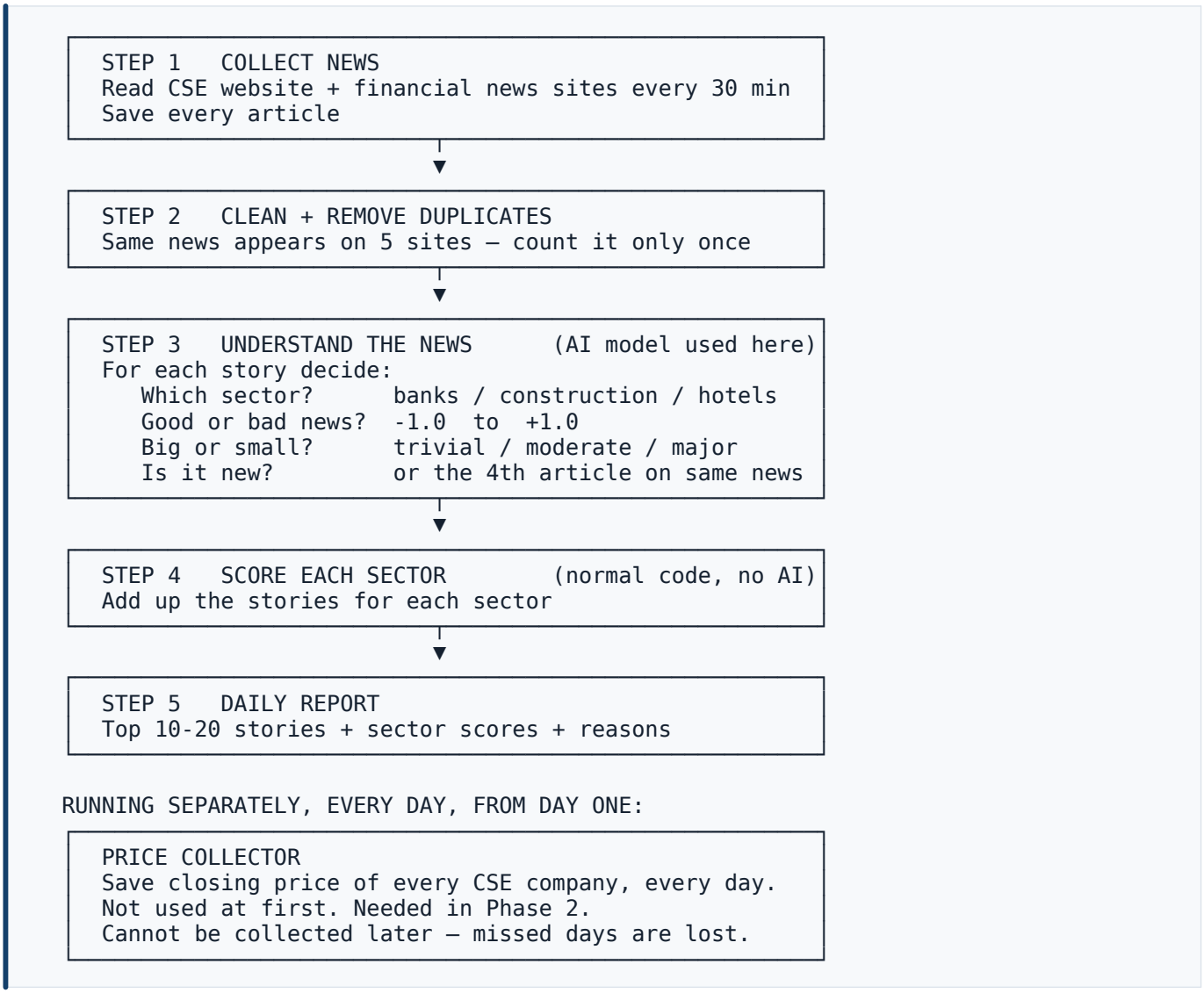
We work at **sector level** instead. We only need to know the story is about *a* bank, not *which* bank. This is much easier and, on a small market like ours, it is also **more reliable**: many CSE shares trade very rarely, so a single share's price moves randomly. Averaging across a whole sector cancels out that random movement.

So this decision makes the project both simpler and better.

3 How the System Works

The system runs five steps in a fixed order, once per day. A sixth job runs separately.

FIGURE 1 — THE DAILY PROCESS



4 What the Output Looks Like

This is the most important page for understanding the project. The system reports evidence and reasons — never instructions.

FIGURE 2 — EXAMPLE OUTPUT, AND WHAT IS FORBIDDEN

```
WHAT THE SYSTEM PRINTS                  WHAT IT MUST NEVER PRINT
──────────────────────                  ────────────────────────

Banks — positive  (+0.62)               "Buy Sampath Bank"
                                        "Price will rise 4%"
In favour:                              "Guaranteed profit"
  3 companies reported
  higher profits                        Reason: the market cannot
  Interest rate was cut                 be predicted. If we promise
                                        numbers we will be wrong
Against:                                often and the tool becomes
  Bank margins shrink                   useless.
  when rates fall
  Sector already up 8%                  We show evidence.
  this week                             The user decides.

How sure: MEDIUM
Based on: 4 news stories
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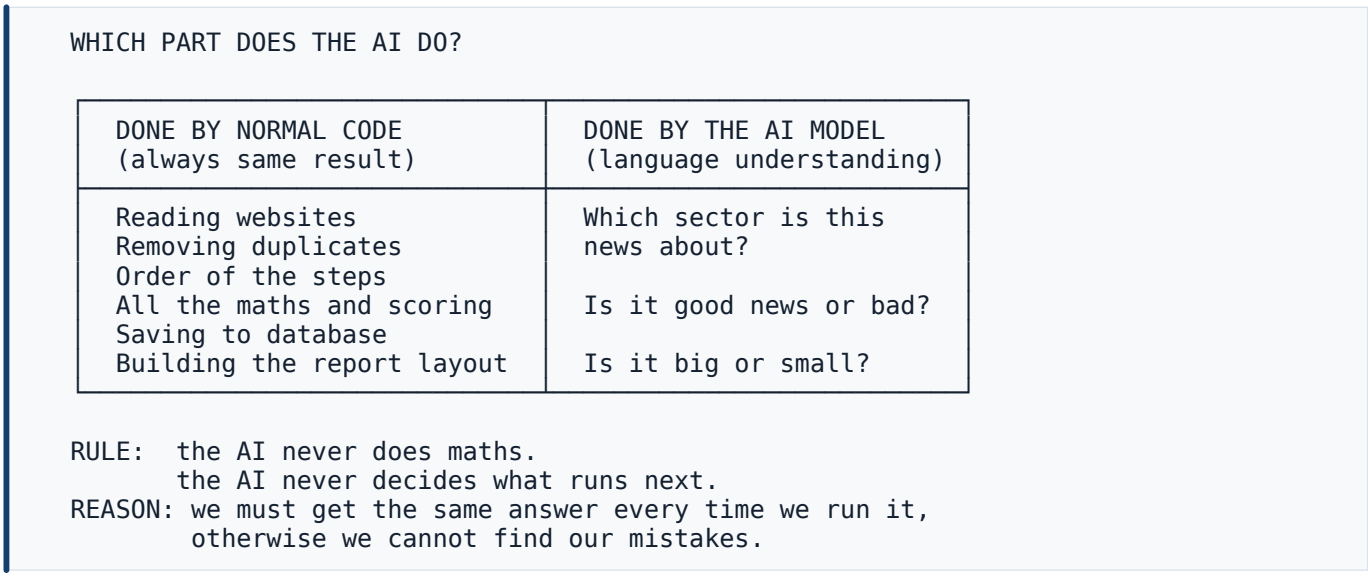
Notice that the report always shows the **arguments against** as well as the arguments in favour, and always states how confident it is. This is intentional. A user can argue with a report like this; a user cannot argue with the word “buy”.

5 System Structure

5.1 Which part does the AI do?

A common mistake in projects like this is to let the AI model do everything. We divide the work clearly.

FIGURE 3 — DIVISION OF WORK



5.2 Main components

COMPONENT	WHAT IT DOES	USES AI?
News collector	Reads news websites and the CSE announcements page every 30 minutes and saves each article.	No
Cleaner	Removes menus, advertisements and HTML so only the article text remains.	No
Duplicate remover	Groups articles that report the same story so it is counted once.	No
Classifier	Reads each story and decides the sector, whether the news is good or bad, and how important it is.	**Yes**
Scorer	Adds up the story scores for each sector using simple arithmetic.	No
Report builder	Arranges the results into the daily report.	No
Price collector	Saves the closing price of every CSE company each day. Runs independently of everything else.	No

6 Code Structure

FIGURE 4 — FOLDER LAYOUT

```
cse-news-agent/
│
├── data/
│   ├── sector_map.csv        company -> sector (made by hand once)
│   ├── sources.yaml          list of news sites to read
│   └── cse.db                database (articles, prices, scores)
│
├── src/
│   ├── collect/
│   │   ├── news_fetcher.py       read news sites
│   │   └── price_fetcher.py      save daily closing prices
│   │
│   ├── process/
│   │   ├── cleaner.py            remove ads, menus, HTML
│   │   └── deduplicator.py       group repeated stories
│   │
│   ├── analyse/
│   │   ├── schema.py             the answer format for the AI
│   │   ├── classifier.py         ask AI: sector? good/bad? size?
│   │   └── scorer.py             add up scores (plain maths)
│   │
│   ├── report/
│   │   └── brief_builder.py      build the daily report
│   │
│   └── pipeline.py               runs steps 1-5 in order
│
├── tests/                        checks that code works
├── notebooks/                    experiments and accuracy checks
├── docs/                         this document, diagrams
└── README.md
```

THE ONE THING WE MUST START IMMEDIATELY

The **price collector** is a small script — perhaps one hour of work — and it does nothing useful on the first day. But it must be running from day one.

The reason: Phase 2 answers the question *“what usually happened after this kind of news?”* That question can only be answered if we already have months of saved prices. We can improve our code at any time, but we can never go back and collect a price from a day we did not save. **Every day we delay is a day permanently lost.**

7 Technology

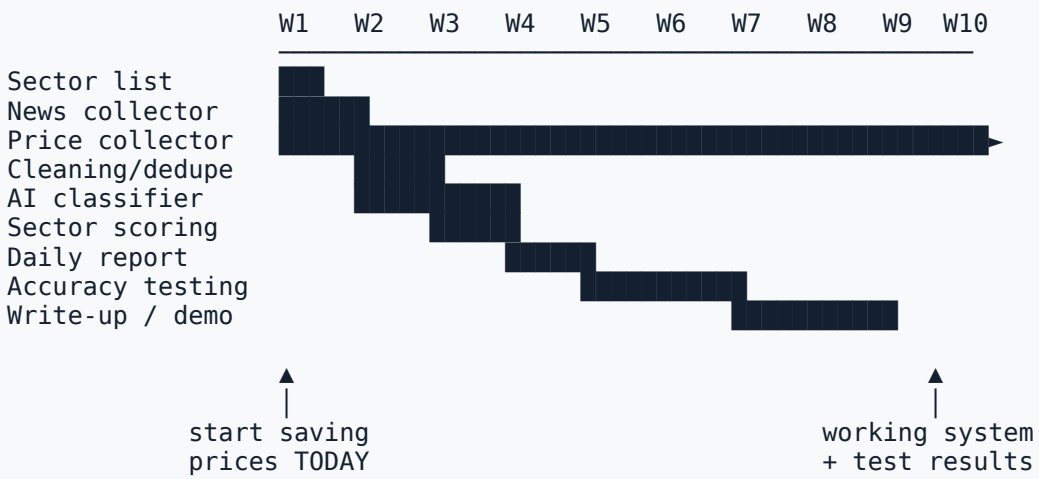
LAYER	TOOL WE PLAN TO USE	WHY
Language	Python	Standard for this type of work; team already knows it.
Pipeline	Plain Python functions, then LangGraph	The steps always run in the same order, so we do not need a complicated agent framework. We will add LangGraph for retries and error handling once the basic version works.
AI model	A large language model with structured output	Used only for reading and judging text. We force it to answer in a fixed format so the output is predictable.
Database	SQLite, then PostgreSQL if needed	SQLite is enough for one machine and needs no setup. Easy to upgrade later.
Web reading	requests + BeautifulSoup, RSS feeds first	RSS is the simplest and most reliable way to get news. We only write custom scrapers where a site has no RSS.
Testing	pytest + hand-labelled article set	The hand-labelled set is how we measure accuracy honestly.

A NOTE ON FRAMEWORKS

We are deliberately starting simple. Our daily process is a straight line — collect, clean, classify, score, report — with no decisions to make along the way. For this, ordinary Python functions are clearer and easier to debug than an agent framework. We will introduce LangGraph when we genuinely need what it provides: automatic retries when a website is down, and saving progress so a failed run can resume.

8 Timeline

FIGURE 5 — TEN-WEEK PLAN



WEEKS	MILESTONE	WHAT WE WILL SHOW
1	Sector list + price collector running	A CSV mapping every CSE company to a sector, and prices saving automatically each day.
2–3	News collection working	Articles from at least four sources arriving and being stored.
3–4	Cleaning and duplicate removal	The same story from five websites correctly grouped into one.
4–6	AI classifier working	Each story labelled with sector, good/bad, and importance.
6–7	Sector scoring + first report	A complete daily report generated automatically.
7–9	Accuracy testing	Measured results against 200 hand-labelled articles (see Section 9).
9–10	Documentation and demonstration	Final report, live demonstration, and honest discussion of limitations.

9 How We Will Measure Success

We will not claim the project works because the output “looks good”. We will measure it.

FIGURE 6 — OUR FOUR TESTS

```
HOW WE PROVE IT WORKS

TEST 1   Is the sector correct?
──────
We label 200 news articles by hand.
We let the system label the same 200.
We compare.
Target: 85 correct out of 100.

TEST 2   Is good/bad correct?
──────
Same 200 articles, checked by 2 team members.
Target: agrees with humans 8 times out of 10.

TEST 3   Does it miss news?
──────
Take one week of CSE announcements.
Check how many the system found.
Target: finds 95 out of 100.

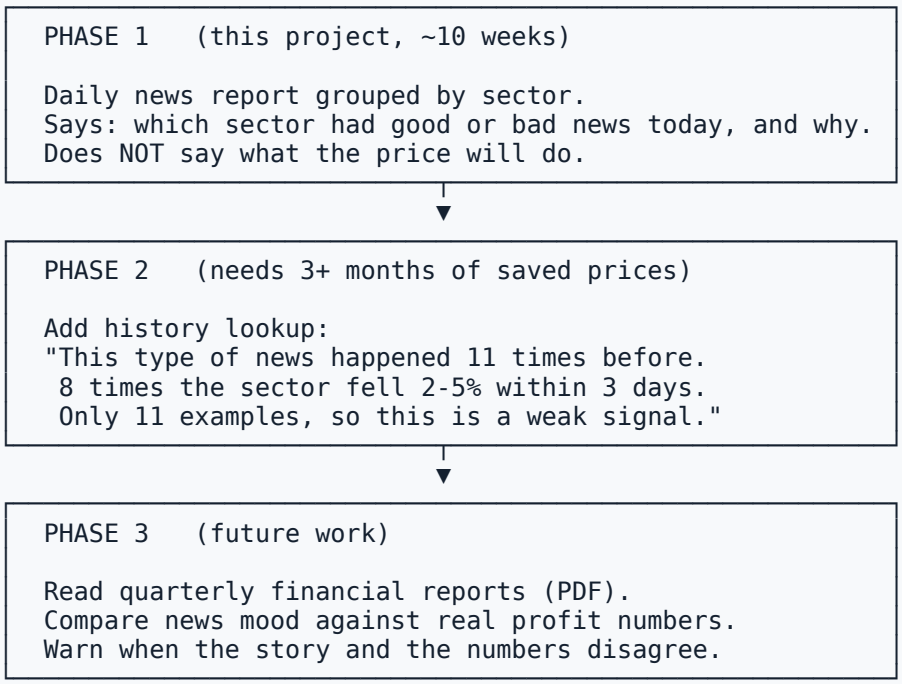
TEST 4   Is it honest about being unsure?
──────
When it says "high confidence", is it usually right?
When it says "low confidence", is it often wrong?
A tool that knows when it is unsure is more useful
than a tool that is slightly more accurate but
always sounds certain.
```

WHY TEST 4 MATTERS MOST

A system that is right 55% of the time *and knows when it is unsure* is more useful than one that is right 60% of the time but always sounds certain. The first can be trusted carefully; the second misleads people. We will report this honestly, including cases where our system performed poorly.

10 Future Work

FIGURE 7 — THE THREE PHASES



11 Risks and Limitations

We list these openly because we would rather plan for them than be surprised.

RISK / LIMITATION	HOW WE HANDLE IT
Websites change their layout and our collector breaks	A daily check that alerts us if any source suddenly returns zero articles, instead of us noticing weeks later.
Some news is in Sinhala or Tamil	Phase 1 handles English sources. We will add translation later, and we will state clearly in our results which sources we covered.
The AI model may label a story wrongly	We require it to give a one-line reason for every answer, so mistakes can be found. We also measure accuracy against 200 hand-labelled articles.
A single large company may dominate its whole sector's score	We limit how much any one company can contribute to a sector score.
Some news affects the whole market, not one sector	Interest rates, exchange rates and budget news are handled as a separate “market-wide” category rather than forced into one sector.
Users may treat the output as investment advice	Every report states its confidence level, always shows arguments against, and carries a clear notice that it is an information tool and not investment advice.
Not enough historical data for Phase 2	This is exactly why we begin saving prices in week 1 rather than when Phase 2 starts.

12 Team and Responsibilities

AREA	RESPONSIBILITY
Data collection	News collector, price collector, source monitoring, database setup.
Processing	Cleaning, duplicate removal, sector mapping.
AI / analysis	Classifier design, prompt and output format, scoring logic.
Evaluation	Hand-labelling the 200-article test set, measuring accuracy, writing results.
Documentation	Project report, diagrams, demonstration.

The hand-labelling work in the evaluation row should be shared by at least two team members, so that we can also measure how often two humans disagree with each other. That number tells us what accuracy is realistically achievable — if two people only agree 85% of the time, we cannot expect the system to do much better.

SUMMARY FOR THE SUPERVISOR

We are building a daily news-analysis tool for the Colombo Stock Exchange that reports which market sectors received good or bad news, with reasons and a confidence level. We have deliberately chosen sector-level analysis over company-level analysis because it is both simpler and more statistically reliable on a small market. We do not attempt price prediction, and we explain why in Section 1.3. Success will be measured against a hand-labelled set of 200 articles, and we will report our weaknesses as well as our strengths.